

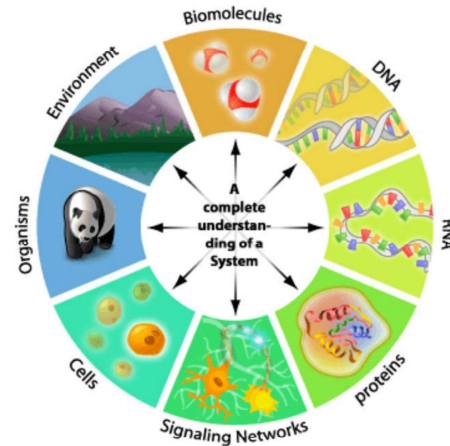
Biological Databases and Resources

Learning Objectives

- Introduction to Databases and Resources
 - Understand how bioinformatics data is stored and organised
 - Describe the different types of data found at the NCBI and EBI resources
 - Locate key bioinformatics databases and resources

Learning Outcomes

- Introduction to Databases and Resources
 - Understand the structure and layout of the NCBI and EBI data resources
 - Understand the difference between databases, tools, repositories
 - Search for data from specific databases using accessions numbers, gene name
 - Use selected tools at NCBI and EBI



Data



Introduction

- Range of different online databases and resources
- Need to know which:
 - Which databases and resources exist
 - What tools are available to mine these resources
 - What tools are available to search across resources

Biological databases



Nucleic Acids Research

The 2024 *Nucleic Acids Research* database issue and the online molecular biology database collection

Daniel J Rigden and Xosé M Fernández

Nucleic Acids Research, Volume 52, Issue D1, 5 January 2024, Pages D1–D9, <https://doi.org/10.1093/nar/gkad1173>



Volume 52, Issue D1
5 January 2024
Cover image
ISSN 0305-1048
EISSN 1362-4962

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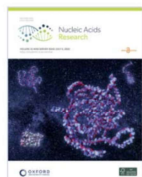
NAR Database Summary Paper Category List

Nucleotide Sequence Databases
RNA sequence databases
Protein sequence databases
Structure Databases
Genomics Databases (non-vertebrate)
Metabolic and Signaling Pathways
Human and other Vertebrate Genomes
Human Genes and Diseases
Microarray Data and other Gene Expression Databases
Proteomics Resources
Other Molecular Biology Databases
Organelle databases
Plant databases
Immunological databases
Cell biology

Editorial: the 21st annual *Nucleic Acids Research* Web Server Issue 2023

Dominik Seelow

Nucleic Acids Research, Volume 51, Issue W1, 5 July 2023, Pages W1–W4, <https://doi.org/10.1093/nar/gkad511>



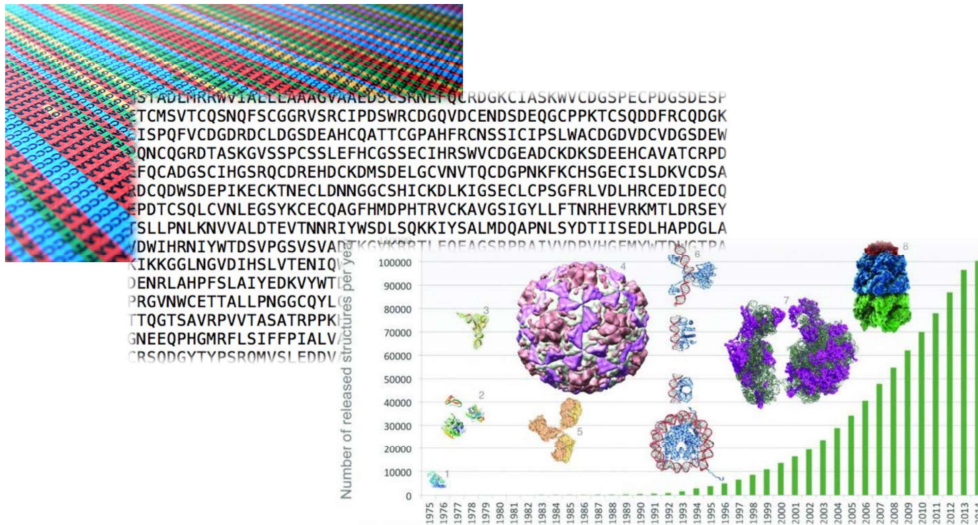
Volume 51, Issue W1
5 July 2023
Cover image
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EISSN 1362-4962

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Databases

- Databases are:
 - Public or private
 - Access and submission
 - Protein, nucleotide, structure, literature, annotation...
 - Generalised or specialised
 - Curated or non-curated
 - Sequence or genome-centred

Primary Databases

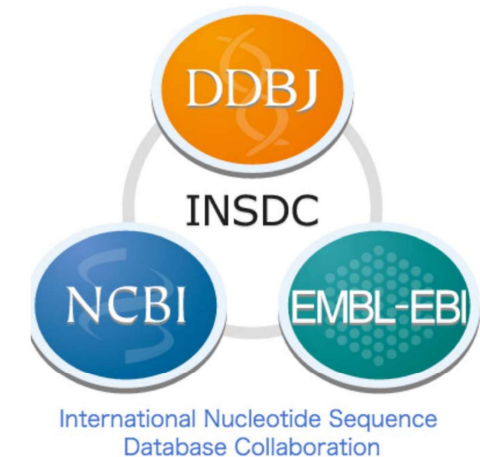


Secondary Databases

- In-depth databases built upon primary sequence data
- Provide several different resources and annotations

Primary Databases

- International Nucleotide Sequence Database Collaboration (INSDC)
- Genomic sequence data stored in 3 public databases
- Each have own accession numbers and tools



Most Popular Bioinformatics Resources

- National Centre for Biotechnology Information (NCBI)
- European Bioinformatics Institute (EMBL-EBI)



NCBI

- National Centre for Biotechnology Information (NCBI)
 - National Institute of Health funded initiative established to store molecular biology information
 - Has grown dramatically since the completion of the human genome project and reduction in sequencing costs
 - Developed and maintain a variety of databases and resources

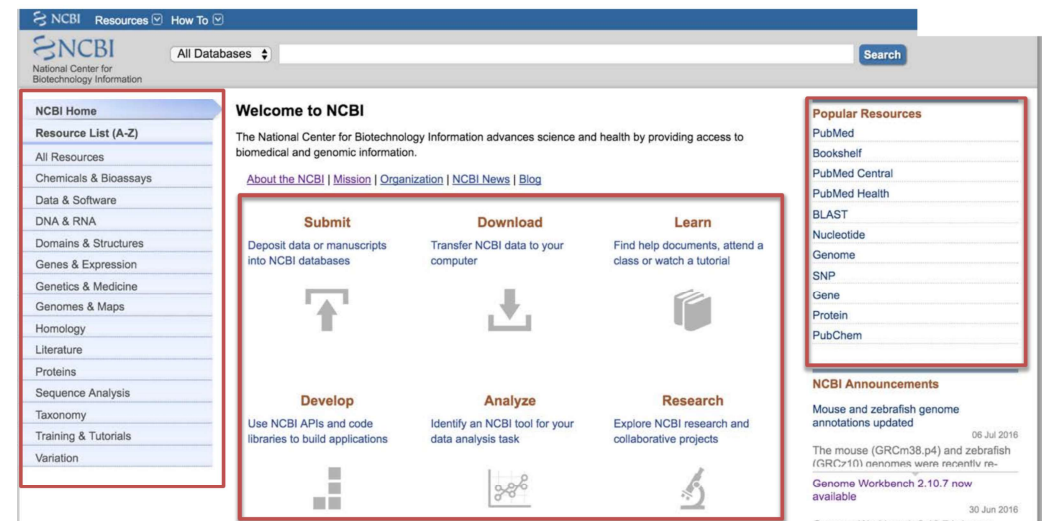
GenBank

- The NIH genetic sequence database
 - Contains an annotated collection of all publicly available DNA sequences
 - Part of INSDC
 - The database is updated on a regular basis, approximately every two months
 - Several divisions within GenBank

GenBank Divisions

- Complete Microbial Genomes
- Whole Genome Shotgun Sequences(WGS)
- Transcriptome Shotgun Assembly Sequences (TSA)
- High-throughout Genomic Sequences (HTGs)
- Targeted Locus Study(TLS)
- Third Party Annotation (TPA)

NCBI



Simple Analysis Tools

Analyze

NCBI provides a wide variety of data analysis tools that allow users to manipulate, align, visualize and evaluate biological data.

Selected Analysis Tools

All Tools	Literature	Health	Genomes	Genes	Proteins	Chemicals
Filter this table						
Tools	Description					
1000 Genomes Browser	Graphically depicts variant calls, genotype calls and supporting evidence (such as aligned sequence reads) that have been produced by the 1000 Genomes Project					
Assembly Archive	Links the raw sequence information found in the Trace Archive with assembly information found in GenBank/EMBL/DBJ					
Basic Local Alignment Search Tool (BLAST)	Finds regions of local similarity between biological sequences					
BLAST Microbial Genomes	Finds regions of local similarity between query sequences and sequences from complete microbial genomes					
Electronic PCR (e-PCR)	Identifies sequence tagged sites (STSs) within DNA sequences					
Gene Plot	Displays pairs of protein homologs that are symmetrical best hits between two genomes					
Genome BLAST	Finds regions of local similarity between query sequences and genome sequences					
Genome ProtMap	Maps each protein from a COG or VOG back to its genome					

Tutorials

Learn

NCBI creates a variety of educational products including courses, workshops, webinars, training materials and documentation. NCBI educational events are free and open to everyone. All NCBI educational materials are available for anyone to re-use and distribute.



Webinars & Courses

In-person courses, live webinars and webinar recordings



Conferences & Presentations

Booth exhibits and workshops at scientific conferences



Tutorials

Tutorials: Training materials in HTML, PDF and video formats



Documentation

Online manuals, handbooks, fact sheets and FAQs



UPCOMING EVENTS

How to upload and analyze dbGaP data in the Cloud

FEBRUARY 3, 2016

Online Webinar: 1:00-2:00pm

Five ways to submit next-gen sequence data to NCBI's Sequence Read Archive

FEBRUARY 17, 2016

Online Webinar: 1:00-2:00pm

"NCBI Resources for Patent Searchers" at the PIUG Biotechnology 2016 Conference

FEBRUARY 24, 2016

Workshop

A Librarian's Guide to NCBI

MARCH 7-11, 2016

Workshop

Experimental Biology 2016 Annual Meeting

NCBI – DNA and RNA

NCBI Home

Resource List (A-Z)

All Resources

Chemicals & Bioassays

Data & Software

DNA & RNA

Domains & Structures

Genes & Expression

Genetics & Medicine

Genomes & Maps

Homology

Literature

Proteins

Sequence Analysis

Taxonomy

Training & Tutorials

Variation

DNA & RNA

All Databases Downloads Submissions Tools How To

Databases

[Assembly](#)
A database providing information on the structure of assembled genomes, assembly names and other meta-data, statistical reports, and links to genomic sequence data.

[BioProject \(formerly Genome Project\)](#)
A collection of genomics, functional genomics, and genetics studies and links to their resulting datasets. This resource describes project scope, material, and objectives and provides a mechanism to retrieve datasets that are often difficult to find due to inconsistent annotation, multiple independent submissions, and the varied nature of diverse data types which are often stored in different databases.

[BioSample](#)
The BioSample database contains descriptions of biological source materials used in experimental assays.

[Consensus CDS \(CCDS\)](#)
A collaborative effort to identify a core set of human and mouse protein coding regions that are consistently annotated and of high quality.

[Database of Expressed Sequence Tags \(dbEST\)](#)
A division of GenBank that contains short single-pass reads of cDNA (transcript) sequences. dbEST can be searched directly through the Nucleotide EST Database.

[Database of Genome Survey Sequences \(dbGSS\)](#)
A division of GenBank that contains short single-pass reads of genomic DNA. dbGSS can be searched directly through the Nucleotide GSS Database.

Quick Links

[BioProject \(formerly Genome Project\)](#)

[Database of Short Genetic Variations \(dbSNP\)](#)

[GenBank](#)

[Nucleotide Database](#)

[PopSet](#)

[RefSeqGene](#)

[Reference Sequence \(RefSeq\)](#)

[Sequence Read Archive \(SRA\)](#)

[Trace Archive](#)

[UniGene](#)

[BLAST \(Stand-alone\)](#)

[GenBank: Bankit](#)

[GenBank: Sequin](#)

[GenBank: tbl2asn](#)

[Basic Local Alignment Search Tool \(BLAST\)](#)

[E-Utilities](#)

[Genome Workbench](#)

[Primer-BLAST](#)

[ProSplign](#)

[Splign](#)

NCBI – Not only DNA data

NCBI Home

Resource List (A-Z)

All Resources

Chemicals & Bioassays

Data & Software

DNA & RNA

Domains & Structures

Genes & Expression

Genetics & Medicine

Genomes & Maps

Homology

Literature

Proteins

Sequence Analysis

Taxonomy

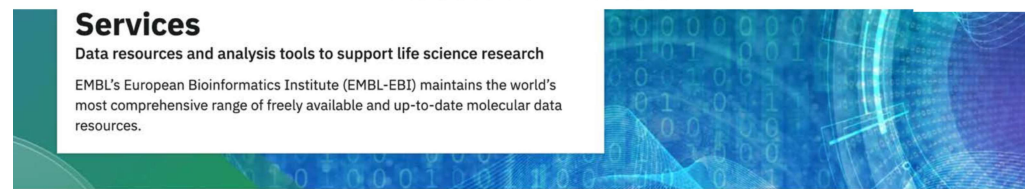
Training & Tutorials

Variation

EMBL - EBI

- Maintain the world's most comprehensive range of freely available and up-to-date molecular databases
- Offer online and live training events for using their resources
 - <https://www.ebi.ac.uk/training/>

EMBL – EBI



Services
Data resources and analysis tools to support life science research

EMBL's European Bioinformatics Institute (EMBL-EBI) maintains the world's most comprehensive range of freely available and up-to-date molecular data resources.

Find a data resource or a tool **Search**

[Explore all our data resources and tools](#) ➔

Our full range of data resources and data analysis tools are essential for supporting life science research.

Featured data resources



AlphaFold DB
Database for protein structure predictions for numerous species
[CC-BY](#)



ArrayExpress
A database of functional genomics experiments, including microarray and RNAseq expression data typically related to publications.
[Web API](#)



BioImage Archive
The BioImage Archive is EMBL-EBI's general purpose image archive, accepting molecular biology imaging data associated with peer-reviewed publications.

EMBL – EBI

EMBL-EBI data resources and tools

EMBL's European Bioinformatics Institute maintains the world's most comprehensive range of freely available and up-to-date molecular data resources.

Find a data resource or a tool **Search**

Showing **1 - 15** out of **157** results

Sort by: Featured ▼



AlphaFold DB
Database for protein structure predictions for numerous species
[DATA RESOURCE](#) [CC-BY](#)



Annotation Platform

Refine by

Type

- ☐ Data resources
- ☐ Tools

Category

- ☐ Chemical biology

ACCESSING DATA

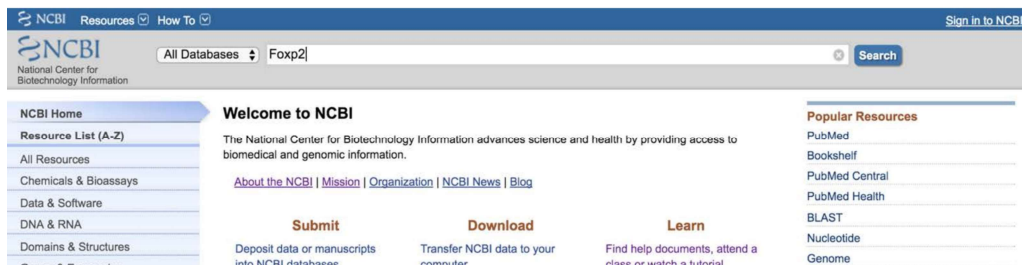
Accessing Data

- Why would you need to access sequence data?
 - Need to know the sequence of a functional regions
 - Identify variants (changes) in a sequence
 - Compare your sequence of interest to those that have been identified previously
 - Find diseases associated with variation in your gene of interest
 - Find sequence datasets from a previously published study

Accessing Data

- Important to be clear what data you are searching for
- Most tools have been developed to link to all annotations for a particular query
- Both NCBI and EBI provide portals to allow you to search across all available databases with a single query

Example: FOXP2 Human



NCBI Portal Search

Results found in 28 databases (1 error)

GENE
FOXP2 – forkhead box P2
Homo sapiens (human)
Also known as: CAGH44, SPCH1, TNRC10
Gene ID: 93986

RefSeq transcripts (8) RefSeq proteins (6) RefSeqGene (1) PubMed (181)

Orthologs Genome Data Viewer BLAST

RefSeq transcripts
FOXP2 – 3 of 8 transcripts

Transcript	Isoform	Len (nt)
NR_033766.2	non-coding	6,355
NR_033767.2	non-coding	6,601
NM_148898.4	II	6,618

[View full table](#) NCBI Datasets

RefSeq Sequences +

Literature

Bookshelf	80
MeSH	5
NLM Catalog	3
PubMed	790
PubMed Central	4,970

Genes

Gene	1,045
GEO DataSets	304
GEO Profiles	11,251
HomoloGene	error
PopSet	13

Proteins

Conserved Domains	3
Identical Protein Groups	181
Protein	4,212
Protein Family Models	5
Structure	3

Genomes

Assembly	0
BioCollections	0
BioProject	46

Clinical

ClinicalTrials.gov	1
ClinVar	395
dbGaP	1

PubChem

BioAssays	0
Compounds	0
Pathways	0

Popular Databases

- **Genes** – One stop resource for all annotation information for a gene
- **PubMed** – Extensive biomedical literature database
- **Nucleotide** – Database of all DNA sequence data
- **dbSNP** – Database of single nucleotide polymorphisms
- **Protein** – Database of protein sequences

Popular Databases

- **RefSeq** – Comprehensive, integrated, well-annotated set of reference sequences – genomic, transcript and protein
- **OMIM** – Online Mendelian Inheritance in Man - Database of human genes and genetic phenotypes
- **ClinVar** – Database of genomic variation and the relationship to human health

Gene Database – Foxp2

Gene: [Search](#)

Gene sources: Genomic, Categories: Alternatively spliced, Annotated genes, Non-coding, Protein-coding, Sequence content, CCDS, Ensembl, RefSeq, RefSeqGene, Status: Current, [Clear all](#), [Show additional filters](#)

Tabular - 20 per page - Sort by Relevance - Send to: [Hide sidebar >>](#)

Filters: [Manage Filters](#)

Results by taxon

Top Organisms: [\[Tree\]](#)
Homo sapiens (95)
Mus musculus (21)
Rattus norvegicus (11)
Salvelinus namaycush (7)
Oncorhynchus mykiss (7)
All other taxa (215)
More...

Find related data
Database: [Select](#)
[Find name](#)

Search details
foxp2[All Fields] AND alive[prop]
[Search](#) [See more...](#)

Recent activity
foxp2 AND alive[prop] (1045)
[Turn Off](#) [Clear](#)

Name/Gene ID	Description	Location	Aliases	MIM
☐ FOXP2 ID: 114142	forkhead box P2 [Mus musculus (house mouse)]	Chromosome 6, NC_000072.7 (14901348..15441976)	2810043D05Rik, CAG-16, D9Kic7	
☐ FOXP2 ID: 93986	forkhead box P2 [Homo sapiens (human)]	Chromosome 7, NC_000007.14 (114086327..114693765)	CAGH44, SPCH1, TNRC10	605317

Search results
Items: 1 to 20 of 1045
[See also 6 discontinued or replaced items.](#)
[First](#) [Previous](#) [Page 1](#) of 53 [Next](#) [Last](#)

Gene Database – Foxp2

FOX2 forkhead box P2 [Homo sapiens (human)]

Gene ID: 93986, updated on 1-Mar-2024

[Download](#)

Summary

Genomic context

Genomic regions, transcripts, and products

Expression

Bibliography

Phenotypes

Variation

HIV-1 interactions

Pathways from PubChem

Interactions

General gene information

General protein information

NCBI Reference Sequences (RefSeq)

Related sequences

Additional links

[Gene LinkOut](#)

Gene Database – Foxp2

FOXP2 forkhead box P2 [*Homo sapiens* (human)]

Gene ID: 93986, updated on 1-Mar-2024

Download Datasets

Summary

Official Symbol FOXP2 provided by HGNC
Official Full Name forkhead box P2 provided by HGNC
Primary source [HGNC:13875](#)
See related [Ensembl:ENSG00000128573](#) [MIM:605317](#); [AllianceGenome:HGNC:13875](#)
Gene type protein coding
RefSeq status REVIEWED
Organism [Homo sapiens](#)
Lineage Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorhini; Catarrhini; Hominidae; Homo
Also known as SPCH1; CAGH44; TNRC10
Summary This gene encodes a member of the forkhead/winged-helix (FOX) family of transcription factors. It is expressed in fetal and adult brain as well as in several other organs such as the lung and gut. The protein product contains a FOX DNA-binding domain and a large polyglutamine tract and is an evolutionarily conserved transcription factor, which may bind directly to approximately 300 to 400 gene promoters in the human genome to regulate the expression of a variety of genes. This gene is required for proper development of speech and language regions of the brain during embryogenesis, and may be involved in a variety of biological pathways and cascades that may ultimately influence language development. Mutations in this gene cause speech-language disorder 1 (SPCH1), also known as autosomal dominant speech and language disorder with orofacial dyspraxia. Multiple alternative transcripts encoding different isoforms have been identified in this gene.[provided by RefSeq, Feb 2010]
Expression Ubiquitous expression in ovary (RPKM 3.6), endometrium (RPKM 3.4) and 22 other tissues [See more](#)
Orthologs [mouse](#) [all](#)
NEW [Try the new Gene table](#)
[Try the new Transcript table](#)

Gene Database – Foxp2

Bibliography

Related articles in PubMed

1. [Human-specific changes in two functional enhancers of FOXP2.](#)
Benitez-Burraco A, et al. Cell Mol Biol (Noisy-le-grand), 2022 Nov 30. PMID 37114314
2. [Functional dissection of two amino acid substitutions unique to the human FOXP2 protein.](#)
Bornschein U, et al. Sci Rep, 2023 Mar 6. PMID 36879029. [Free PMC Article](#)
3. [Investigation of the forkhead box protein P2 gene by the next-generation sequence analysis method in children diagnosed with specific learning disorder.](#)
Yazici M, et al. Psychiatr Genet, 2023 Feb 1. PMID 36617742
4. [In-depth characterisation of a cohort of individuals with missense and loss-of-function variants disrupting FOXP2.](#)
Morison LD, et al. J Med Genet, 2023 Jun. PMID 36328423. [Free PMC Article](#)
5. [Do variants in the coding regions of FOXP2, a gene implicated in speech disorder, confer a risk for congenital amusia?](#)
Peretz I, et al. Ann N Y Acad Sci, 2022 Nov. PMID 35980667

[See all \(181\) citations in PubMed](#)

GeneRIFs: Gene References Into Functions

[What's a GeneRIF?](#)

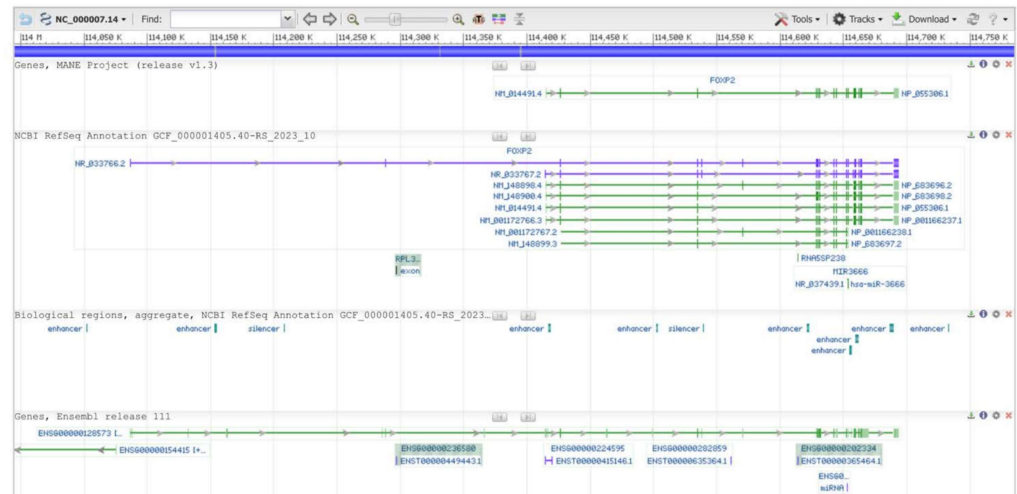
Gene Database – Foxp2

Genomic regions, transcripts, and products

[Go to reference sequence details](#)

Genomic Sequence: [NC_000007.14](#) Chromosome 7 Reference GRCh38.p14 Primary Assembly ▾

[Go to nucleotide](#); [Graphics](#); [FASTA](#); [GenBank](#)



Gene Database – Foxp2

Phenotypes

[BioGRID CRISPR Screen Phenotypes \(18 hits/1281 screens\)](#)

[Find tests for this gene in the NIH Genetic Testing Registry \(GTR\)](#)

[Review eQTL and phenotype association data in this region using PheGenI](#)

Associated conditions

Description	Tests
Childhood apraxia of speech	
MedGen: C0750927 , OMIM: 602081 , GeneReviews: FOXP2-Related Speech and Language Disorder	Compare labs

Copy number response

Description
Copy number response
Triplosensitivity
No evidence available (Last evaluated 2012-09-19)
ClinGen Genome Curation Page
Haploinsufficiency
Little evidence for dosage pathogenicity (Last evaluated 2012-09-19)
ClinGen Genome Curation Page , PubMed

Gene Database – Foxp2

Phenotypes

[BioGRID CRISPR Screen Phenotypes \(18 hits/1281 screens\)](#)
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Copy number response

Description
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Triplosensitivity No evidence available (Last evaluated 2012-09-19) ClinGen Genome Curation Page
Haploinsufficiency Little evidence for dosage pathogenicity (Last evaluated 2012-09-19) ClinGen Genome Curation Page , PubMed

Gene Database– Foxp2

NCBI Reference Sequences (RefSeq)

[NEW Try the new Transcript table](#)

RefSeqs maintained independently of Annotated Genomes

These reference sequences exist independently of genome builds. [Explain](#)

Genomic

1. **NG_007491.3 RefSeqGene**

Range: 333506..612456
Download: [GenBank](#), [FASTA](#), [Sequence Viewer \(Graphics\)](#)

mRNA and Protein(s)

1. **NM_001172766.3 → NP_001166237.1 forkhead box protein P2 isoform V**
[See identical proteins and their annotated locations for NP_001166237.1](#)

Status: REVIEWED

Description	Transcript Variant: This variant (5) lacks an in-frame exon and uses an alternate in-frame splice site in the coding region, compared to variant 2. The resulting isoform (V) is shorter than isoform II.
Source sequence(s)	AC020606 , AC073626 , AF337817 , AI369947 , BC018016 , BC143867
UniProtKB/TrEMBL	A0A0U1RCM2
Related	ENSP00000377135.2 , ENST00000393498.6
Conserved Domains (2) summary	
cd00059	FH; Forkhead (FH), also known as a "winged helix". FH is named for the Drosophila fork head protein, a transcription factor which promotes terminal rather than segmental development. This family of transcription factor domains, which bind to B-DNA as ...
Location:503 → 575	
pfam16159	FOXP-CC; FOXP coiled-coil domain
Location:341 → 408	

Gene Database– Foxp2

Transcripts and Proteins

FOXP2 – forkhead box P2

Homo sapiens (human)

Download ▾ Select columns 8 Transcripts 8 selected							
Gene ID	Gene symbol	Transcript	Length (nt)	Protein	Length (aa)	Protein name	Isofo
93986	FOXP2	NR_033766.2	6355				
93986	FOXP2	NR_033767.2	6601				
93986	FOXP2	NM_148898.4	6618	NP_683696.2	740	forkhead box protein P2	II
93986	FOXP2	NM_148900.4	6594	NP_683698.2	732	forkhead box protein P2	IV
93986	FOXP2	NM_014491.4	6543	NP_055306.1	715	forkhead box protein P2	I
93986	FOXP2	NM_001172766.3	6540	NP_001166237.1	714	forkhead box protein P2	V
93986	FOXP2	NM_001172767.2	1485	NP_001166238.1	457	forkhead box protein P2	VI
93986	FOXP2	NM_148899.3	1410	NP_683697.2	432	forkhead box protein P2	III

Gene Database– Foxp2

NCBI Reference Sequences (RefSeq)

[NEW Try the new Transcript table](#)

RefSeqs maintained independently of Annotated Genomes

These reference sequences exist independently of genome builds. [Explain](#)

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1. **NG_007491.3 RefSeqGene**

Range: 333506..612456
Download: [GenBank](#), [FASTA](#), [Sequence Viewer \(Graphics\)](#)

mRNA and Protein(s)

1. **NM_001172766.3 → NP_001166237.1 forkhead box protein P2 isoform V**
[See identical proteins and their annotated locations for NP_001166237.1](#)

Status: REVIEWED

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UniProtKB/TrEMBL	A0A0U1RCM2
Related	ENSP00000377135.2 , ENST00000393498.6
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Location:503 → 575	
pfam16159	FOXP-CC; FOXP coiled-coil domain
Location:341 → 408	

GenBank Entry – Foxp2

Send to: ▼

```

FEATURES
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CDS
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229871..229975,237389..237476,239112..239195,
243252..243453,244542..244618,244758..244859,
247251..247372,248636..248705,249459..249622,
274968..278951)

```

☒ Complete Record
☐ Coding Sequences
☐ Gene Features

Choose Destination

☐ File ☐ Clipboard
☐ Collections ☐ Analysis

- Each GenBank record, consisting of both a sequence and its annotations is assigned a unique identifier called an accession number

LOCUS	NG_007491	278951 bp	DNA	linear	PRI 11-FEB-2024
DEFINITION	Homo sapiens forkhead box P2 (FOXP2), RefSeqGene on chromosome 7.				
ACCESSION	NG_007491 REGION: 333506..612456				
VERSION	NG_007491.3				
KEYWORDS	RefSeq; RefSeqGene.				
SOURCE	Homo sapiens (human)				
ORGANISM	Homo sapiens				
	Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;				

Accession Number Prefixes

Accession prefixes	Type	Description
NC_, NG_	Known Refseq	Genomic regions or assembly
NM_	Known Refseq	mRNA
NR_	Known Refseq	RNA
NP_	Known Refseq	Protein
NT_, NW_	Known Refseq	Genomic contig or scaffold
XM_	Model Refseq	mRNA
XR_	Model Refseq	RNA
XP_	Model Refseq	Protein

EMBL-EBI

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EMBL's European Bioinformatics Institute

EMBL-EBI

Unleashing the potential of big data in biology

Please fill in this field

Find a gene, protein or chemical

All

Search

Example searches: blast keratin bfl1 | About EBI Search

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First datasets available in the



DECIPHER v11.24 Released



We are EMBL: Jacqueline



Europe PMC supports preprint

EMBL –

Source

All results (9,607)

Search results for **foxp2**

Showing 42 results out of 9,607 in All results

Give us feedback on these results

Filter your results

Source

All results (9,607)

- Genomes & metagenomes (418)
- Nucleotide sequences (1,495)
- Protein sequences (4,896)
- Macromolecular structures (860)
- Gene expression (86)
- Diseases (27)
- Molecular interactions (49)
- Gene-Disease Associations (350)
- Reactions & pathways (6)
- Protein families (27)
- Protein expression data (2)
- Literature (1,052)
- Samples & ontologies (327)
- EMBL-EBI website (1)
- Other data (11)

Gene and protein summary

Forkhead Box P2
FOXP2 (ENSG00000128573)
Human (Homo sapiens)

Forkhead Box P2
Foxp2 (ENSMUSG00000029563)
House Mouse (Mus musculus)

Protein sequences (4,896 result)

Source: UniProtKB (ID: FOXP2_HUMAN)

O15409

Forkhead box protein P2
Homo sapiens(Reviewed)
Secondary accession number(s): A0AUV6, A4D

Genomes & metagenomes (418)

Nucleotide sequences (1,495)

Protein sequences (4,896)

Macromolecular structures (860)

Gene expression (86)

Diseases (27)

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Other data (11)

NFO1.08NF02.

EMBL – EBI – Foxp2

Gene and protein summary for foxp2

Organism human (Homo sapie

Gene Expression Protein Protein Structure Literature

forkhead box P2

View in Ensembl

Gene information and sequence

FOXP2 spans 607446 bps of chromosome 7 from 114086327 to 114693772.

FOXP2 has 32 transcripts containing a total of 112 exons on the forward strand.

Annotation for this gene includes both automatic annotation from Ensembl and Havana manual curation, see article.

View the gene sequence in Ensembl.

View the chromosome region for this gene in Ensembl.

Variations

FOXP2 has 224345 SNPs.

View sequence variations such as polymorphisms, along with genotypes and disease associations in Ensembl.

Orthologues

FOXP2 has 209 orthologues in Ensembl

View homology between species inferred from a gene tree in Ensembl.

Paralogues

FOXP2 has 4 paralogues in Ensembl

EMBL – EBI – Foxp2

Gene-based displays

- Summary
 - Splice variants
 - Transcript comparison
 - Gene alleles
- Sequence
 - Secondary Structure
- Comparative Genomics
 - Genomic alignments
 - Gene tree
 - Gene path-loss tree
 - Orthologues
 - Paralogues
- Ontologies
 - GO: Molecular function
 - GO: Biological process
 - GO: Cellular component
- Phenotypes
 - Genetic Variation
 - Variant table
 - Variant image
 - Structural variants
- Gene expression
 - Pathways
 - Molecular interactions
- Other assemblies
 - Regulation
 - External references
 - Supporting evidence
 - ID History
 - Gene history

Gene: FOXP2 ENSG00000128573

Description
forward bias P2 [Source:HGNC Symbol;Acc:[HGNC:138754](#)]

Gene Synonyms
CAQH44, SPCH1, TNRC10

Location
Chromosome 7: 114,086,327-114,693,772 forward strand.
GRCh38:CM000669.2

About this gene
This gene has 32 transcripts ([splice variants](#)), 209 orthologues, 5 paralogs and is associated with 4 phenotypes.

Transcripts
[Show transcript table](#)

Summary ⓘ

Name
FOXP2^g (HGNC Symbol)

MANE
This gene contains MANE Select [ENST00000359098](#), [ENSP00000296438](#)

UniProtKB
This gene has proteins that correspond to the following UniProtKB identifiers: [Q1A509^f](#)

RefSeq
This Ensembl/Gencode gene contains transcripts(s) for which we have [selected identical RefSeq transcript\(s\)](#). If there are other RefSeq transcripts available they will be in the [External references](#) table

CDS
This gene is a member of the Human CDS set: [CCDS84935.1^f](#), [CCDS95154.1^f](#), [CCDS8760.1^f](#), [CCDS5781.2^f](#)

Ensembl version
ENSG00000128573.28

Other assemblies
This gene maps to [113,728,382-114,333,627](#) in GRCh37 coordinates.
View this locus in the GRCh37 archive: [ENSG00000128573^f](#)

Gene type
Protein coding

Annotation method
Annotation for this gene includes both automatic annotation from Ensembl and Havana manual curation, see [article](#).

Go to Region in Detail for more tracks and navigation options (e.g. zooming!)

Addmore tracks | Custom tracks | Share | Resize image | Export image | Reset configuration | Reset track order |

Genes: [Compendio](#)
set from GENCODE v41

The figure shows a genomic browser view of the FOXP2 gene region. The x-axis represents the genomic position from 114,136 to 114,714. Four tracks are displayed, each representing a different transcript isoform:

- FOXP2-213 - ENST00000412402**: non-ret mediated decay
- FOXP2-212 - ENST00000404349**: non-ret mediated decay
- FOXP2-225 - ENST00000703012**: protein coding
- FOXP2-228 - ENST00000735014**: protein coding

The tracks show exons as colored boxes (green for non-ret mediated decay, red for protein coding) and introns as lines with arrows indicating the direction of transcription. The top track is labeled "Forward strand".

EMBL – EBI – FoxP2

[illegible]

EMBL – EBI – FoxP2

xxx Gene Expression **Protein** Protein Structure Literature

Forkhead box protein P2

[View In UniProt](#) 

Function

Transcriptional repressor that may play a role in the specification and differentiation of lung epithelium. May also play a role in developing neural, gastrointestinal and cardiovascular tissues. Can act with CTPB1 to synergistically repress transcription but CTPBP1 is not essential. Plays a role in synapse formation by regulating SRPX2 levels. Involved in neural mechanisms mediating the development of speech and language.

[View annotation in UniProt](#)

Subcellular Location

Nucleus.

[View annotation in UniProt](#)

Disease

Speech-language disorder 1 (SPCH1) [MIM:602081]: A disorder characterized by severe orofacial dyspraxia resulting in largely incomprehensible speech. Affected individuals have severe impairment in the selection and sequencing of fine orofacial movements which are necessary for articulation, and deficits in several facets of grammatical skills and language processing, such as the ability to break up words into their constituent phonemes. Note—The disease is caused by variants affecting the gene represented in this entry.

Note=A chromosomal aberration involving FOXP2 is a cause of severe speech and language impairment.
Translocation t(5;7)(q22;q31.2).

[View annotation in UniProt](#)

EMBL – EBI – FoxP2

xxx Gene Expression Protein Protein Structure Literature

Crystal Structure of Foxp2 bound Specifically to DNA.

[View In PDBe](#)

Method

X-ray diffraction

Experiment

- Resolution: 1.9Å
- R-Factor: 23.11%
- Free R-Factor: 24.36%

Dates

- Deposited: 16-06-2005
- Released: 31-01-2006
- Revised: 14-02-2024

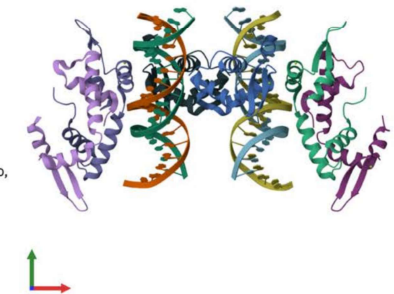
Deposited by

Stroud, J.C., Wu, Y., Bates, D.L., Han, A., Nowick, K., Paabo, S., Tong, H., Chen, L.

Primary citation

Structure of the forkhead domain of FOXP2 bound to DNA.

Structure vol:14 page:159-66 (2006)

[View citation in PDB](#)

Ribbon structure of 2a07

EMBL – EBI – FoxP2

Gene Expression Protein Protein Structure Literature

All Articles (4241)

PMID: 38418803 • PMCID:

Novel FOXP2 variant associated with speech and language dysfunction in a Chinese family and literature review.

Che F, Li C, Zhang L, Qian C, Mo L, Li B, Wu H, Wang L, Yang Y.
J Appl Genet[2024]
Cited: 0 times

PMID: 38414006 • PMCID: PMC10897999

Emerging roles of circular RNAs in tumorigenesis, progression, and treatment of gastric cancer.

Ma Q, Yang F, Xiao B, Guo X.
J Transl Med[2024, 22(1):207]
Cited: 0 times

PMID: 38383296 • PMCID: PMC10880338

Impact of insomnia upon inflammatory digestive diseases and biomarkers: a two-sample mendelian randomization research on Europeans.

Dai L, Ye Y, Mugaanyi J, Lu C, Lu C.
BMC Gastroenterol[2024, 24(1):79]
Cited: 0 times

Other popular resources at EBI

- **Ensembl** – resource for high quality integrated annotation data
- **Uniprot** – Universal Protein Resource for protein sequence and functional annotation data
- **PDB** – Protein data bank Europe – Collection of 3D structural data
- **InterPro** – database of protein families, domains and conserved sites

Specialised Databases

- A large number of specialised databases exist
 - Most of the sequences are also in GenBank/EMBL/DDBJ
 - Usually disease or organism specific
 - May integrate data not found in other resources i.e. clinical information + genetic data
 - Specific analysis and visualization tools for working with the datasets

Specialised Databases

The screenshot displays the PlasmoDB website, a specialized database for Plasmodium informatics. The header features the PlasmoDB logo, a search bar with the placeholder text "Site search, e.g. PF3D7_1135400 or 'lactate dehydrogenase'", and navigation links for "My Strategies", "Searches", "Tools", "My Workspace", "Data", "About", "Help", and "Contact Us". Below the header, the main content area is titled "Overview of Resources and Tools" and includes a "Getting Started" section. The "Getting Started" section provides instructions on how to use the site search and explore data. The search bar in the main content area is labeled "Search for..." and includes a "Filter the searches below..." option. The search results are categorized into "Genes", "Organisms", "Popset Isolate Sequences", "Genomic Segments", "SNPs", and "SNPs (from Array)".

Specialised Databases

Archived Type Search

Data sets All types Keyword

Data directory



Human

1000 Genomes

The 1000 Genomes Project developed a new map of the human genome at a resolution that was unmatched by other ...



Bacterial

Bacterial Data

This Archived page provides historical information on the genome sequences of bacteria that were sequenced at the Wellcome Sanger Institute and ...



Bacteriophage

Bacteriophage Data

All bacteriophage genomes sequenced by the Sanger Institute



Human

Cancer Gene Census

The *Cancer Gene Census* is a high-confidence list of genes with substantial published evidence in oncology.



Bacterial

Culture collection of human gut bacteria

Host-Microbiota Interactions Laboratory



Bacterial

Culture collection of reference Clostridium difficile isolates

Host-Microbiota Interactions Laboratory

Specialised Databases

GISAID About us Database Features Events Collaborations Resources Help Login

In Focus

16 million hCoV-19 sequences shared: Where do surveillance efforts stand?

After mixed efforts in 2020, followed by a massive explosion of Omicron data generated by late 2021 from 215 countries and territories, the number of unique submitting countries, and hence, representativeness of global surveillance, today remains at early 2021 levels, and well above 2020 levels.

While the US, UK, and Germany contributed an outsized share in 2022, it is worth pointing out that increasing representation of diverse regions over time, and the timeliness of submissions are far more critical to ensure new variants are detected in real-time. The recent timely discovery of BA.2.86 and EG.5.1 are examples of continued effectiveness so far.

中國新冠疫情 Lineage comparison

hCoV-19 data sharing via GISAID

16,586,033
genome sequence submissions

Specialised Databases

GWAS Catalog

The NHGRI-EBI Catalog of human genome-wide association studies

Search the catalog

Examples: Parkinson disease, rs3093017, Yao, 2q37.2, HBS1L, 6:167120000-167130000, GCST90132222, PMID:35241825

Download

Download a full copy of the GWAS Catalog in spreadsheet format as well as current and older versions of the GWAS diagram in SVG format.

Summary statistics

Documentation and access to full summary statistics for GWAS Catalog studies where available.

Submit

Submit summary statistics to GWAS Catalog.

Documentation

Including FAQs, our curation process, training materials, related resources, a list of abbreviations and API documentation.

Diagram

Explore an interactive visualisation of all SNP-trait associations with genome-wide significance ($p \leq 5 \times 10^{-8}$).

Ancestry

An introduction to our ancestry curation process.

Specialised Databases

European Genome-phenome Archive

Quick tour

What is EGA?

The European Genome-Phenome Archive (EGA) is a collaborative partnership between the [European Bioinformatics Institute \(EMBL-EBI\)](#) and the [Centre for Genomic Regulation \(CRG\)](#).

The EGA provides a service for the permanent archiving and distribution of personally identifiable genetic and phenotypic data resulting from biomedical research projects. Data at EGA was collected from individuals whose consent agreements authorise data release only for specific research use to bona fide researchers. Strict protocols govern how information is managed, stored, and distributed by the EGA project.

Why do we need the EGA?

Participants in medical or genetic research projects typically provide consent for their data to be used in research but not for open public distribution. These data require a secure archiving, processing, and disseminating service that respects the original informed consent agreements. In recent years, governments worldwide have enacted data privacy protection laws and regulations to protect the rights of their citizens, further restricting how personal data is shared. In this environment, services for securely archiving and sharing sensitive human data for

Course overview

Search within this course

- What is EGA?**
 - What can I do with the EGA?
- Searching for data in the EGA
- Accessing the data in the EGA
- Submitting data to the EGA
- Summary

Your feedback

- Get help and support on the EGA

Summary

- Databases are used to store different types of biological data
- Primary databases store raw sequence data
- Secondary databases provide information on the annotation of the sequence data
- Knowledge of what databases exists makes it easier to find specific information
- NCBI and EBI are the two most popular resources for extracting biological data and contain several different databases
- Portals are useful for retrieving information from several different resources with a single query
- Specialised databases/resources are useful for finding organism/disease or method specific data